

## SOFTWARE DEMONSTRATION - DATABASE AND SOFTWARE DEVELOPMENT

### **“HOPLIAS”: AN ACCESSIBLE WEB TOOLKIT FOR AUTOMATED KARYOTYPE ASSEMBLY**

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Cytogenetics plays a fundamental role in understanding genomic structural and evolutionary processes. The study of chromosomes, their structures, rearrangements and abnormalities, and organization in genome provides crucial information for numerous applications, from biomedical research and precision medicine to studies surrounding the diversification of our biodiversity. However, the assembly of chromosomes into a karyotype, especially in non-model organisms is a complex and time-consuming task, requiring specialized technical skills and extensive software knowledge. Chromosome segmentation software's are usually unable to correctly isolate the chromosomes from the image of a non-model organism metaphase. Hybrid platforms that merge manual and automatized tasks are usually expensive for researchers from developing countries. In this scenario, we developed the “Hoplias”, an experimental web application for automatic and supervised karyotypes assembly of fish, that recognizes the elements in the microscopy photograph of the metaphase plate (nuclei, residues, chromosomes), cuts, classifies according to arms ratio and aligns the chromosomes into a karyotype, following the standard classification for fishes based on the arm ratios. Using Support Vector

Machine (SVM) classification algorithm to classify objects between chromosomes and background, we implemented a web platform using modern technologies such as ReactJS and Python. ReactJS was used to create user interfaces accessible on all devices, while a set of APIs based on microservices architecture will ensure efficiency and scalability, using the Nameco framework to generate four microservices: a user control service, an image segmentation service, a chromosome classification service, and a gateway service for observability and security. To perform image filtering we first converted it to grayscale, then we applied the 3x3 mean filter, which removes noise, preserves the contours, and small details of the chromosomes. Mathematical morphology algorithms were used to separate the chromosome pixels from background pixels, improving the quality of the binary image. All these applications were trained and tested upon a dataset of chromosomes of distinct fish species, from a repository provided by the Laboratório de Genética Ecológica e Evolutiva (LaGEEvo - UFV) and Laboratório de Citogenética de Peixes (LCP – UFSCar). This data was divided into 80% images for training and 20% images for testing, then the predictions of the SVM classifier were compared with the classified particles, where the software demonstrated good sensitivity to identify and correctly segmentate the chromosomes at the images. The workflow in platform consists of image selection, followed by automatic verification, correction and isolation of chromosomes. The application also allows correction on the position and limits of chromosomes and their centromeres, turning it suitable for several organisms. The last step is the automatic classification and organization of chromosomes in the karyotype, which also allow manual correction through the editor. Further improvements on the graphical interface, implementation of machine learning methods and training on other non-model organisms, that include species with microchromosomes (birds and reptiles), will still be carried out to expand the applicability of the software to other non-model organisms.

Palavras-chave: cytogenetics; chromosomes; computer vision; image segmentation.